

Class 18: Pertussis and the CMI-PB project

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Background

Pertussis (whooping cough) is a common lung infection caused by the bacteria B. Pertussis. It can infect anyone but is most deadly for infants (under 1 year of age).

CDC Tracking Data

The CDC data

Q1. With the help of the R “addin” package datapasta assign the CDC pertussis case number data to a data frame called cdc and use ggplot to make a plot of cases numbers over time.

```
library(datapasta)

cdc <- data.frame(
  Year = c(1922L,1923L,1924L,1925L,
           1926L,1927L,1928L,1929L,1930L,1931L,
           1932L,1933L,1934L,1935L,1936L,
           1937L,1938L,1939L,1940L,1941L,1942L,
```

```

1943L, 1944L, 1945L, 1946L, 1947L,
1948L, 1949L, 1950L, 1951L, 1952L,
1953L, 1954L, 1955L, 1956L, 1957L, 1958L,
1959L, 1960L, 1961L, 1962L, 1963L,
1964L, 1965L, 1966L, 1967L, 1968L, 1969L,
1970L, 1971L, 1972L, 1973L, 1974L,
1975L, 1976L, 1977L, 1978L, 1979L, 1980L,
1981L, 1982L, 1983L, 1984L, 1985L,
1986L, 1987L, 1988L, 1989L, 1990L,
1991L, 1992L, 1993L, 1994L, 1995L, 1996L,
1997L, 1998L, 1999L, 2000L, 2001L,
2002L, 2003L, 2004L, 2005L, 2006L, 2007L,
2008L, 2009L, 2010L, 2011L, 2012L,
2013L, 2014L, 2015L, 2016L, 2017L, 2018L,
2019L, 2020L, 2021L, 2022L, 2023L, 2024L, 2025L),
Cases = c(107473, 164191, 165418, 152003,
202210, 181411, 161799, 197371,
166914, 172559, 215343, 179135, 265269,
180518, 147237, 214652, 227319, 103188,
183866, 222202, 191383, 191890, 109873,
133792, 109860, 156517, 74715, 69479,
120718, 68687, 45030, 37129, 60886,
62786, 31732, 28295, 32148, 40005,
14809, 11468, 17749, 17135, 13005, 6799,
7717, 9718, 4810, 3285, 4249, 3036,
3287, 1759, 2402, 1738, 1010, 2177, 2063,
1623, 1730, 1248, 1895, 2463, 2276,
3589, 4195, 2823, 3450, 4157, 4570,
2719, 4083, 6586, 4617, 5137, 7796, 6564,
7405, 7298, 7867, 7580, 9771, 11647,
25827, 25616, 15632, 10454, 13278,
16858, 27550, 18719, 48277, 28639, 32971,
20762, 17972, 18975, 15609, 18617,
6124, 2116, 3044, 7063, 21996, 22538 )
)

```

Q. I want a plot of year vs cases

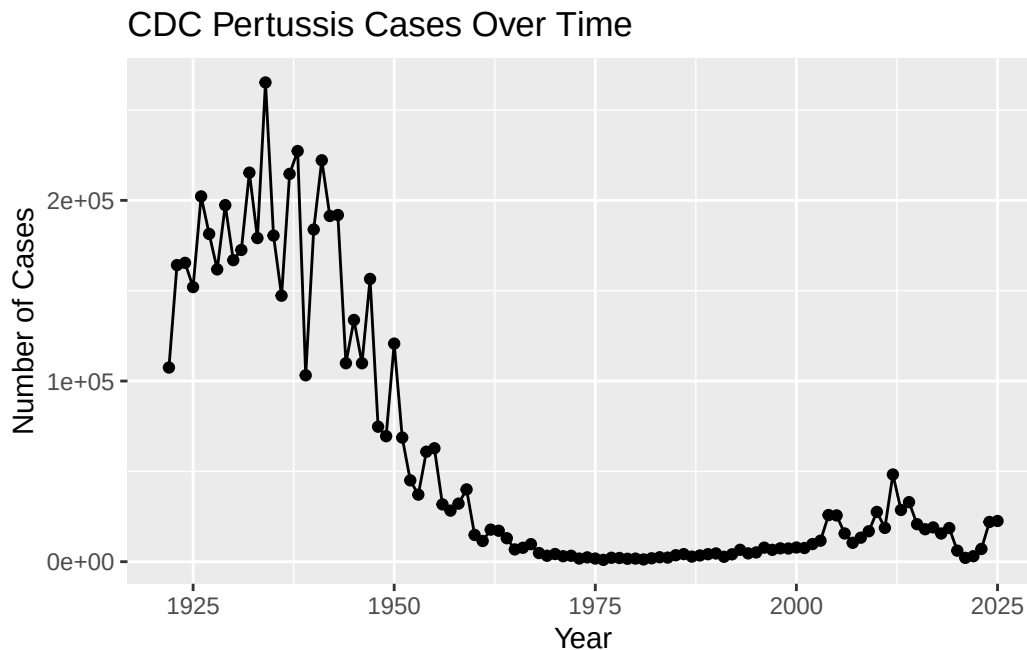
```

library(ggplot2)

ggplot(cdc) +
  aes(x = Year, y = Cases) +

```

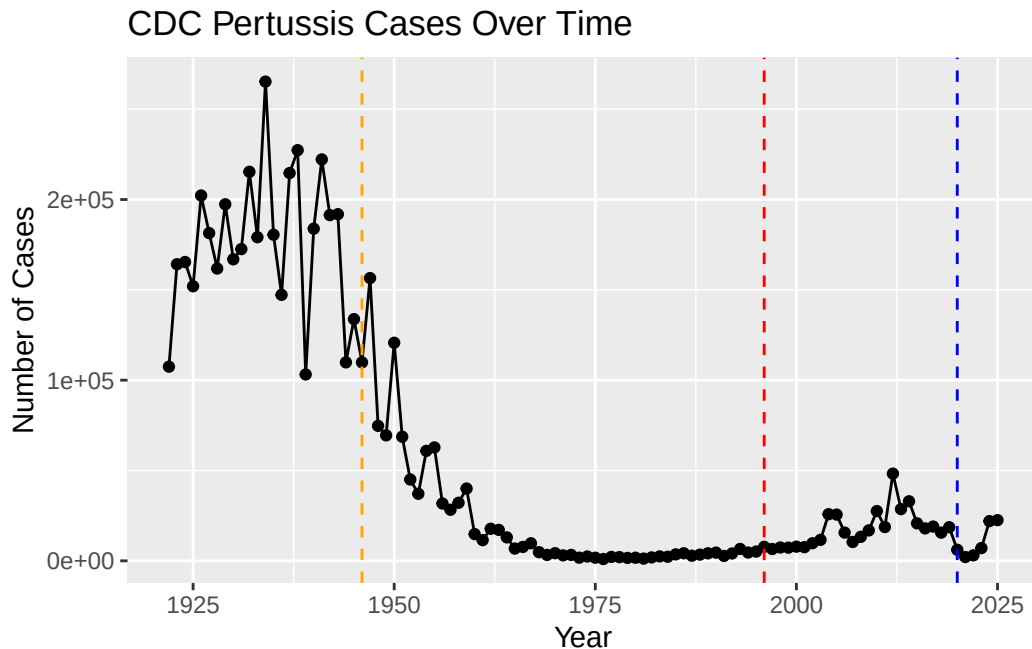
```
geom_point() +
geom_line() +
labs(title = "CDC Pertussis Cases Over Time",
      x = "Year",
      y = "Number of Cases")
```



Q2. Using the ggplot `geom_vline()` function add lines to your previous plot for the 1946 introduction of the wP vaccine and the 1996 switch to aP vaccine (see example in the hint below). What do you notice?

After the introduction of the wP vaccine in 1946, pertussis cases dropped dramatically, but cases began to increase again after the switch to the aP vaccine in 1996.

```
ggplot(cdc) +
  aes(x = Year, y = Cases) +
  geom_point() +
  geom_line() +
  geom_vline(xintercept = 1946, col="orange", lty=2) +
  geom_vline(xintercept = 1996, col="red", lty =2) +
  geom_vline(xintercept = 2020, col="blue", lty =2) +
  labs(title = "CDC Pertussis Cases Over Time",
        x = "Year",
        y = "Number of Cases")
```



Q3. Describe what happened after the introduction of the aP vaccine? Do you have a possible explanation for the observed trend?

After the aP vaccine was introduced, pertussis cases gradually increased, possibly because immunity from the acellular vaccine weakens faster than the original whole-cell vaccine.

Exploring CMI-PB data

The CMI-PB project < <https://www.cmi-pb.org/> >: The mission of CMI-PB is to provide the scientific community with a comprehensive, high-quality and freely accessible resource of Pertussis booster vaccination.

Basically, make available a large dataset on the immune response to Pertussis. They use a “booster” vaccination as a proxy for Pertussis infection.

They make their data available as JSON format API. We can read this into R with the `read_json()` function from the **jsonlite** package.

```
library(jsonlite)
subject <- read_json("https://www.cmi-pb.org/api/subject", simplifyVector = TRUE)
head(subject)
```

	subject_id	infancy_vac	biological_sex	ethnicity	race
1	1	wP	Female	Not Hispanic or Latino	White
2	2	wP	Female	Not Hispanic or Latino	White
3	3	wP	Female	Unknown	White
4	4	wP	Male	Not Hispanic or Latino	Asian
5	5	wP	Male	Not Hispanic or Latino	Asian
6	6	wP	Female	Not Hispanic or Latino	White

	year_of_birth	date_of_boost	dataset
1	1986-01-01	2016-09-12	2020_dataset
2	1968-01-01	2019-01-28	2020_dataset
3	1983-01-01	2016-10-10	2020_dataset
4	1988-01-01	2016-08-29	2020_dataset
5	1991-01-01	2016-08-29	2020_dataset
6	1988-01-01	2016-10-10	2020_dataset

Q4. How many aP and wP infancy vaccinated subjects are in the dataset?

87

```
table(subject$infancy_vac)
```

```
aP wP
87 85
```

Q5. How many Male and Female subjects/patients are in the dataset?

way more females at 112 than males at 60

```
table(subject$biological_sex)
```

```
Female    Male
112      60
```

Q6. What is the breakdown of race and biological sex (e.g. number of Asian females, White males etc...)?

The dataset is mostly composed of White and Asian participants, with 48 White females and 32 White males, and 32 Asian females and 12 Asian males, while other racial groups are represented in much smaller numbers.

Q. is this representative of the US population?

No, it is not representative of the U.S. population because the dataset is dominated by White and Asian participants while other racial groups are underrepresented.

Side-Note: Working with dates

```
library(lubridate)
```

Attaching package: 'lubridate'

The following objects are masked from 'package:base':

date, intersect, setdiff, union

```
today()
```

```
[1] "2026-03-09"
```

```
today() - ymd("2000-01-01")
```

Time difference of 9564 days

```
time_length(today() - ymd("2000-01-01"), "years")
```

```
[1] 26.1848
```

Q7. Using this approach determine (i) the average age of wP individuals, (ii) the average age of aP individuals; and (iii) are they significantly different?

```
subject$age <- today() - ymd(subject$year_of_birth)
```

```
time_length(mean(subject$age[subject$infancy_vac == "wP"]), "years")
```

```
[1] 36.83086
```

```
time_length(mean(subject$age[subject$infancy_vac == "aP"]), "years")
```

```
[1] 28.08015
```

The average age of wP individuals is about 36.8 years while the average age of aP individuals is about 28.1 years, showing that wP individuals are older because the whole-cell vaccine was used earlier.

Q8. Determine the age of all individuals at time of boost?

The ages at the time of booster vaccination vary across individuals but generally range from about 18 to about 50 years old.

```
subject$age_at_boost <- ymd(subject$date_of_boost) - ymd(subject$year_of_birth)
```

```
time_length(subject$age_at_boost, "years")
```

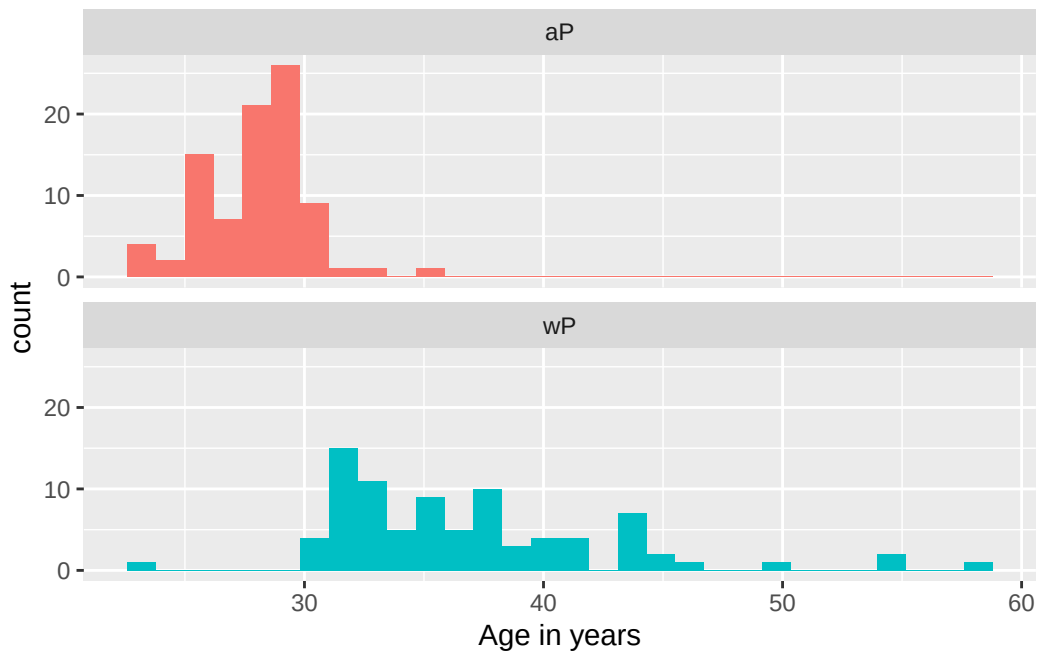
```
[1] 30.69678 51.07461 33.77413 28.65982 25.65914 28.77481 35.84942 34.14921
[9] 20.56400 34.56263 30.65845 34.56263 19.56194 23.61944 27.61944 29.56331
[17] 36.69815 19.65777 22.73511 35.65777 33.65914 31.65777 25.73580 24.70089
[25] 28.70089 33.73580 19.73443 34.73511 19.73443 28.73648 27.73443 19.81109
[33] 26.77344 33.81246 25.77413 19.81109 18.85010 19.81109 31.81109 22.81177
[41] 31.84942 19.84942 18.85010 18.85010 19.90691 18.85010 20.90897 19.04449
[49] 20.04381 19.90691 19.90691 19.00616 19.00616 20.04381 20.04381 20.07940
[57] 21.08145 20.07940 20.07940 20.07940 32.26557 25.90007 23.90144 25.90007
[65] 28.91992 42.92129 47.07461 47.07461 29.07324 21.07324 21.07324 28.15058
[73] 24.15058 24.15058 21.14990 21.14990 31.20876 26.20671 32.20808 27.20876
[81] 26.20671 21.20739 20.26557 22.26420 19.32375 21.32238 19.32375 19.32375
[89] 22.41752 20.41889 21.41821 19.47707 23.47707 20.47639 21.47570 19.47707
[97] 35.90965 28.73648 22.68309 20.83231 18.83368 18.83368 27.68241 32.68172
[105] 27.68241 25.68378 23.68241 26.73785 32.73648 24.73648 25.79603 25.79603
[113] 25.79603 31.79466 19.83299 21.91102 27.90965 24.06297 23.90965 27.12115
[121] 22.12183 23.12115 26.17933 22.17933 29.17728 29.23477 26.23682 28.29295
[129] 31.29363 26.29432 24.35044 27.35113 25.40999 32.41068 27.56194 27.41136
[137] 24.50650 22.56263 29.56057 21.69473 26.69678 31.90691 19.90691 23.90691
[145] 20.90623 31.00616 23.00616 35.00616 32.00548 32.00548 31.04449 28.12047
[153] 25.11978 26.11910 26.19302 22.19302 26.19302 23.19507 29.19370 27.32923
[161] 30.32717 24.55852 30.55715 32.55852 30.55715 22.67488 26.67488 32.67625
[169] 20.67625 31.75086 20.86516 36.06297
```

Q9. With the help of a faceted boxplot or histogram (see below), do you think these two groups are significantly different?

Yes, the two groups appear significantly different because the wP group is generally older while the aP group is mostly younger.

```
ggplot(subject) +
  aes(time_length(age, "year"),
      fill = as.factor(infancy_vac)) +
  geom_histogram(show.legend = FALSE) +
  facet_wrap(vars(infancy_vac), nrow = 2) +
  xlab("Age in years")
```

`stat\_bin()` using `bins = 30`. Pick better value `binwidth`.



Joining multiple tables

```
table(subject$race, subject$biological_sex)
```

	Female	Male
American Indian/Alaska Native	0	1
Asian	32	12
Black or African American	2	3
More Than One Race	15	4

Native Hawaiian or Other Pacific Islander	1	1
Unknown or Not Reported	14	7
White	48	32

We can read more tables from the CMI-PB database

```
speciman <- read_json("http://cmi-pb.org/api/v5_1/specimen", simplifyVector = T)
ab_titer <- read_json("http://cmi-pb.org/api/v5_1/plasma_ab_titer", simplifyVector = T)
```

```
head(speciman)
```

	specimen_id	subject_id	actual_day_relative_to_boost
1	1	1	-3
2	2	1	1
3	3	1	3
4	4	1	7
5	5	1	11
6	6	1	32

	planned_day_relative_to_boost	specimen_type	visit
1	0	Blood	1
2	1	Blood	2
3	3	Blood	3
4	7	Blood	4
5	14	Blood	5
6	30	Blood	6

```
head(ab_titer)
```

	specimen_id	isotype	is_antigen_specific	antigen	MFI	MFI_normalised
1	1	IgE	FALSE	Total	1110.21154	2.493425
2	1	IgE	FALSE	Total	2708.91616	2.493425
3	1	IgG	TRUE	PT	68.56614	3.736992
4	1	IgG	TRUE	PRN	332.12718	2.602350
5	1	IgG	TRUE	FHA	1887.12263	34.050956
6	1	IgE	TRUE	ACT	0.10000	1.000000

	unit	lower_limit_of_detection
1	UG/ML	2.096133
2	IU/ML	29.170000
3	IU/ML	0.530000
4	IU/ML	6.205949
5	IU/ML	4.679535
6	IU/ML	2.816431

To make sense of all this data we need to “join” (a.k.a.”merge” or “link”) all these tables together. only then will uou know that a give Ab measurment was collected on a ceretin data form a certion data from a certin wP or aP subject (from the `subject` table)

we can use **dplyr** and the `*_join()` family of functions to do this.

Q9. Complete the code to join specimen and subject tables to make a new merged data frame containing all specimen records along with their associated subject details:

Yes, the two groups appear significantly different because the wP group is generally older while the aP group is mostly younger.

```
library(dplyr)
```

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

`filter`, `lag`

The following objects are masked from 'package:base':

`intersect`, `setdiff`, `setequal`, `union`

```
meta <- inner_join(subject, speciman)
```

Joining with `by = join\_by(subject\_id)`

```
head(meta)
```

	subject_id	infancy_vac	biological_sex	ethnicity	race
1	1	wP	Female Not	Hispanic or Latino	White
2	1	wP	Female Not	Hispanic or Latino	White
3	1	wP	Female Not	Hispanic or Latino	White
4	1	wP	Female Not	Hispanic or Latino	White
5	1	wP	Female Not	Hispanic or Latino	White
6	1	wP	Female Not	Hispanic or Latino	White

	year_of_birth	date_of_boost	dataset	age	age_at_boost	specimen_id
1	1986-01-01	2016-09-12	2020_dataset	14677 days	11212 days	1

2	1986-01-01	2016-09-12	2020_dataset	14677 days	11212 days	2
3	1986-01-01	2016-09-12	2020_dataset	14677 days	11212 days	3
4	1986-01-01	2016-09-12	2020_dataset	14677 days	11212 days	4
5	1986-01-01	2016-09-12	2020_dataset	14677 days	11212 days	5
6	1986-01-01	2016-09-12	2020_dataset	14677 days	11212 days	6

	actual_day_relative_to_boost	planned_day_relative_to_boost	specimen_type
1	-3	0	Blood
2	1	1	Blood
3	3	3	Blood
4	7	7	Blood
5	11	14	Blood
6	32	30	Blood

	visit
1	1
2	2
3	3
4	4
5	5
6	6

let's do one more `inner_join()` to join the `ab_titer` which all out `meta` data

Q10. Now using the same procedure join `meta` with `titer` data so we can further analyze this data in terms of time of visit aP/wP, male/female etc.

```
abdata <- inner_join(ab_titer, meta)
```

Joining with ``by = join_by(specimen_id)``

```
head(abdata)
```

	specimen_id	isotype	is_antigen_specific	antigen	MFI	MFI_normalised
1	1	IgE	FALSE	Total	1110.21154	2.493425
2	1	IgE	FALSE	Total	2708.91616	2.493425
3	1	IgG	TRUE	PT	68.56614	3.736992
4	1	IgG	TRUE	PRN	332.12718	2.602350
5	1	IgG	TRUE	FHA	1887.12263	34.050956
6	1	IgE	TRUE	ACT	0.10000	1.000000

	unit	lower_limit_of_detection	subject_id	infancy_vac	biological_sex
1	UG/ML	2.096133	1	wP	Female
2	IU/ML	29.170000	1	wP	Female

3	IU/ML	0.530000	1	wP	Female
4	IU/ML	6.205949	1	wP	Female
5	IU/ML	4.679535	1	wP	Female
6	IU/ML	2.816431	1	wP	Female
	ethnicity	race	year_of_birth	date_of_boost	dataset
1	Not Hispanic or Latino	White	1986-01-01	2016-09-12	2020_dataset
2	Not Hispanic or Latino	White	1986-01-01	2016-09-12	2020_dataset
3	Not Hispanic or Latino	White	1986-01-01	2016-09-12	2020_dataset
4	Not Hispanic or Latino	White	1986-01-01	2016-09-12	2020_dataset
5	Not Hispanic or Latino	White	1986-01-01	2016-09-12	2020_dataset
6	Not Hispanic or Latino	White	1986-01-01	2016-09-12	2020_dataset
	age	age_at_boost	actual_day_relative_to_boost		
1	14677 days	11212 days		-3	
2	14677 days	11212 days		-3	
3	14677 days	11212 days		-3	
4	14677 days	11212 days		-3	
5	14677 days	11212 days		-3	
6	14677 days	11212 days		-3	
	planned_day_relative_to_boost	specimen_type	visit		
1		0	Blood	1	
2		0	Blood	1	
3		0	Blood	1	
4		0	Blood	1	
5		0	Blood	1	
6		0	Blood	1	

Q11. How many specimens (i.e. entries in abdata) do we have for each isotype?

Six antibody isotypes are measured in the dataset: IgE (6698), IgG (7265), IgG1 (11993), IgG2 (12000), IgG3 (12000), and IgG4 (12000).

```
table(abdata$isotype)
```

IgE	IgG	IgG1	IgG2	IgG3	IgG4
6698	7265	11993	12000	12000	12000

Q hoe many diffrent 'antigen' values are measured?

There are 16 different antigen values measured in the dataset.

```
table(abdata$antigen)
```

ACT	BETV1	DT	FELD1	FHA	FIM2/3	LOLP1	LOS	Measles	OVA
1970	1970	6318	1970	6712	6318	1970	1970	1970	6318
PD1	PRN	PT	PTM	Total	TT				
1970	6712	6712	1970	788	6318				

Q12. What are the different \$dataset values in abdata and what do you notice about the number of rows for the most “recent” dataset?

There are four dataset values: 2020_dataset, 2021_dataset, 2022_dataset, and 2023_dataset. The most recent datasets contain fewer rows than the earlier datasets, likely because fewer samples have been collected or processed so far.

```
table(abdata$dataset)
```

2020_dataset	2021_dataset	2022_dataset	2023_dataset
31520	8085	7301	15050

Lets focus on IgG istype

```
igg <- abdata |>
  filter(isotype=="IgG")
head(igg)
```

	specimen_id	isotype	is_antigen_specific	antigen	MFI	MFI_normalised
1	1	IgG	TRUE	PT	68.56614	3.736992
2	1	IgG	TRUE	PRN	332.12718	2.602350
3	1	IgG	TRUE	FHA	1887.12263	34.050956
4	19	IgG	TRUE	PT	20.11607	1.096366
5	19	IgG	TRUE	PRN	976.67419	7.652635
6	19	IgG	TRUE	FHA	60.76626	1.096457

	unit	lower_limit_of_detection	subject_id	infancy_vac	biological_sex
1	IU/ML	0.530000	1	wP	Female
2	IU/ML	6.205949	1	wP	Female
3	IU/ML	4.679535	1	wP	Female
4	IU/ML	0.530000	3	wP	Female
5	IU/ML	6.205949	3	wP	Female

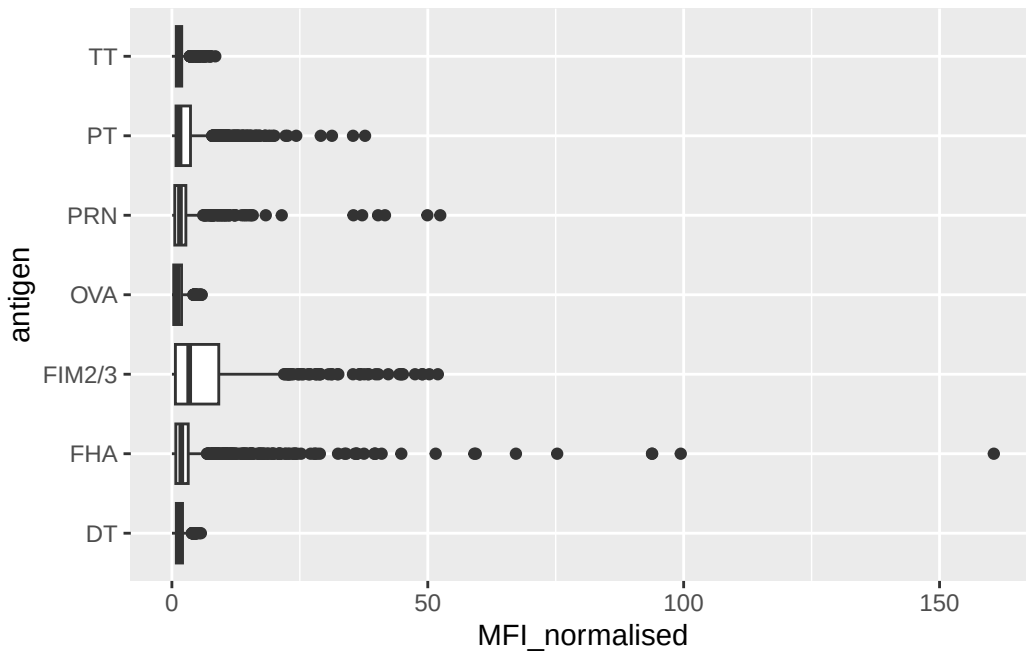
6	IU/ML		4.679535	3	wP	Female
		ethnicity	race	year_of_birth	date_of_boost	dataset
1	Not Hispanic or Latino	White	1986-01-01	2016-09-12	2020_dataset	
2	Not Hispanic or Latino	White	1986-01-01	2016-09-12	2020_dataset	
3	Not Hispanic or Latino	White	1986-01-01	2016-09-12	2020_dataset	
4		Unknown	White	1983-01-01	2016-10-10	2020_dataset
5		Unknown	White	1983-01-01	2016-10-10	2020_dataset
6		Unknown	White	1983-01-01	2016-10-10	2020_dataset
	age	age_at_boost	actual_day_relative_to_boost			
1	14677 days	11212 days		-3		
2	14677 days	11212 days		-3		
3	14677 days	11212 days		-3		
4	15773 days	12336 days		-3		
5	15773 days	12336 days		-3		
6	15773 days	12336 days		-3		
	planned_day_relative_to_boost	specimen_type	visit			
1		0	Blood	1		
2		0	Blood	1		
3		0	Blood	1		
4		0	Blood	1		
5		0	Blood	1		
6		0	Blood	1		

We make a plot of `MFI_normalised` values for all `antigen` values

Q13. Complete the following code to make a summary boxplot of Ab titer levels (MFI) for all antigens:

The antigens PT, FIM2/3, and FHA appear to have the widest range of antibody titer values.

```
ggplot(igg) +
  aes(MFI_normalised, antigen) +
  geom_boxplot()
```



The antigens “PT”, “FIM2/3” and “FHA” appear to have the widest range of values

Q14. What antigens show differences in the level of IgG antibody titers recognizing them over time? Why these and not others?

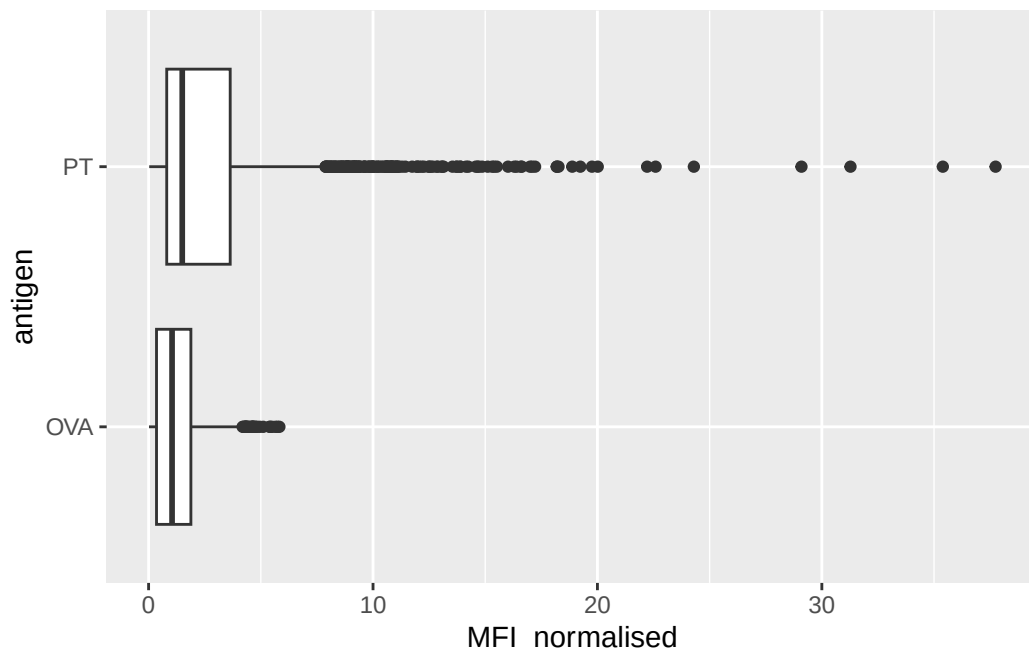
The antigens PT, FHA, FIM2/3, and PRN show the largest differences in IgG antibody titers. These antigens are components of the pertussis vaccine, so they trigger stronger immune responses, while unrelated antigens such as OVA or measles show much smaller changes.

Q15. Filter to pull out only two specific antigens for analysis and create a boxplot for each. You can choose any you like. Below I picked a “control” antigen (“OVA”, that is not in our vaccines) and a clear antigen of interest (“PT”, Pertussis Toxin, one of the key virulence factors produced by the bacterium *B. pertussis*).

Two antigens were selected for analysis: PT, a pertussis toxin antigen found in the vaccine, and OVA, which serves as a control antigen not related to pertussis.

```
pt_ova <- igg |>
  filter(antigen %in% c("PT", "OVA"))
```

```
ggplot(pt_ova) +
  aes(MFI_normalised, antigen) +
  geom_boxplot()
```



For this comparison I selected the control antigen OVA and the pertussis antigen PT. OVA is not part of the pertussis vaccine and serves as a baseline control, while PT (Pertussis toxin) is a major antigen in the vaccine and should show a stronger immune response.

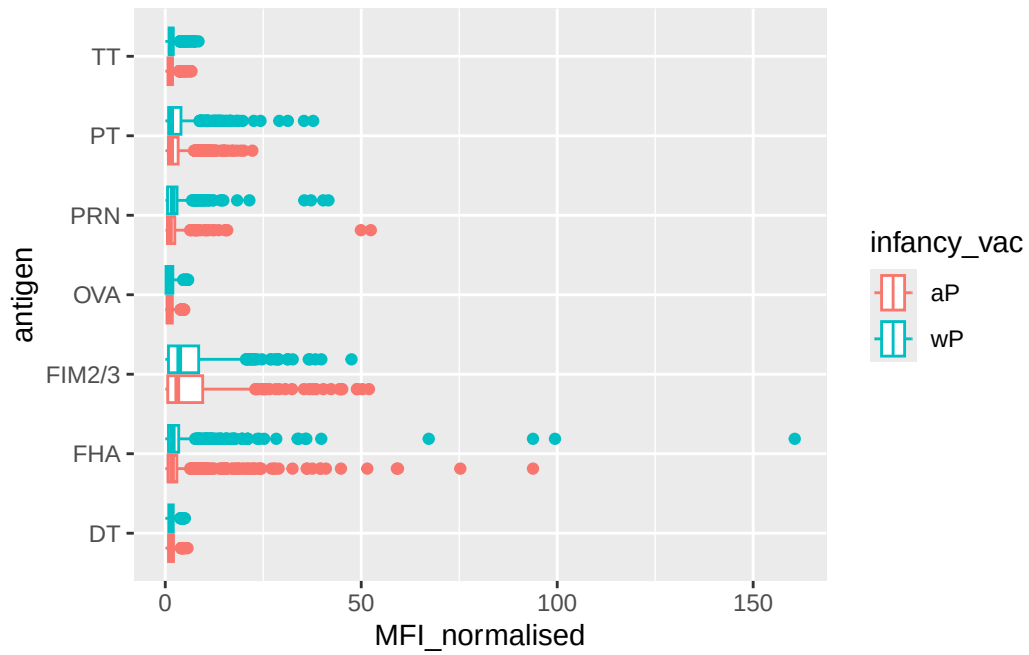
Q16. What do you notice about these two antigens time courses and the PT data in particular?

The PT antigen shows a clear increase in IgG antibody levels after the booster vaccination, while the control antigen OVA shows little change over time because it is not related to the pertussis vaccine.

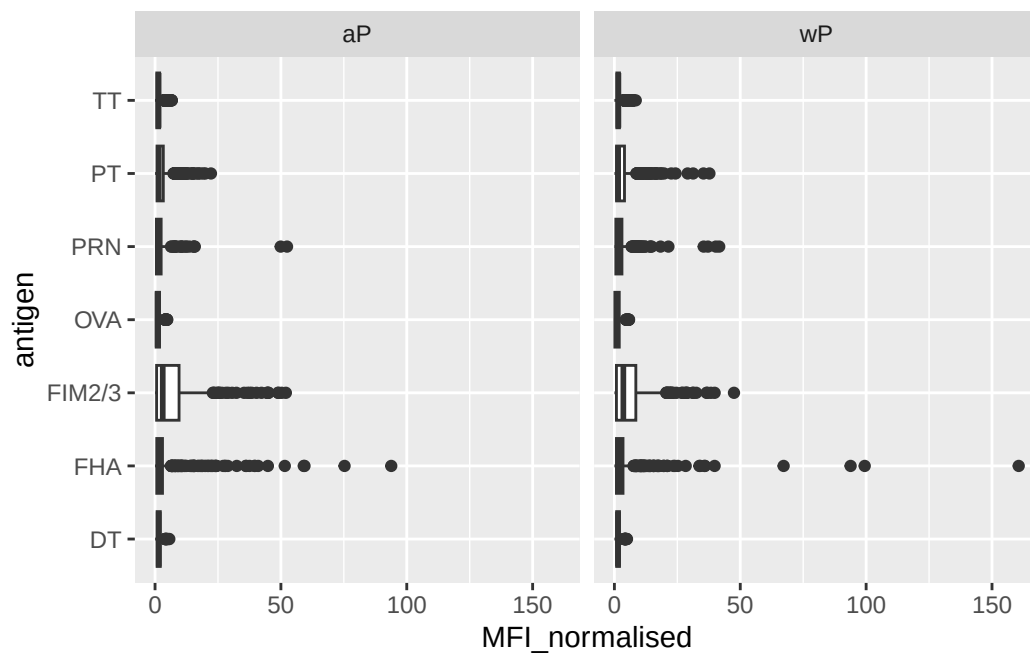
Q17. Do you see any clear difference in aP vs. wP responses?

Yes, there appears to be some differences between aP and wP responses, with wP individuals generally showing slightly stronger antibody responses to several pertussis antigens compared to aP individuals.

```
ggplot(igg) +
  aes(MFI_normalised, antigen, col=infancy_vac) +
  geom_boxplot()
```

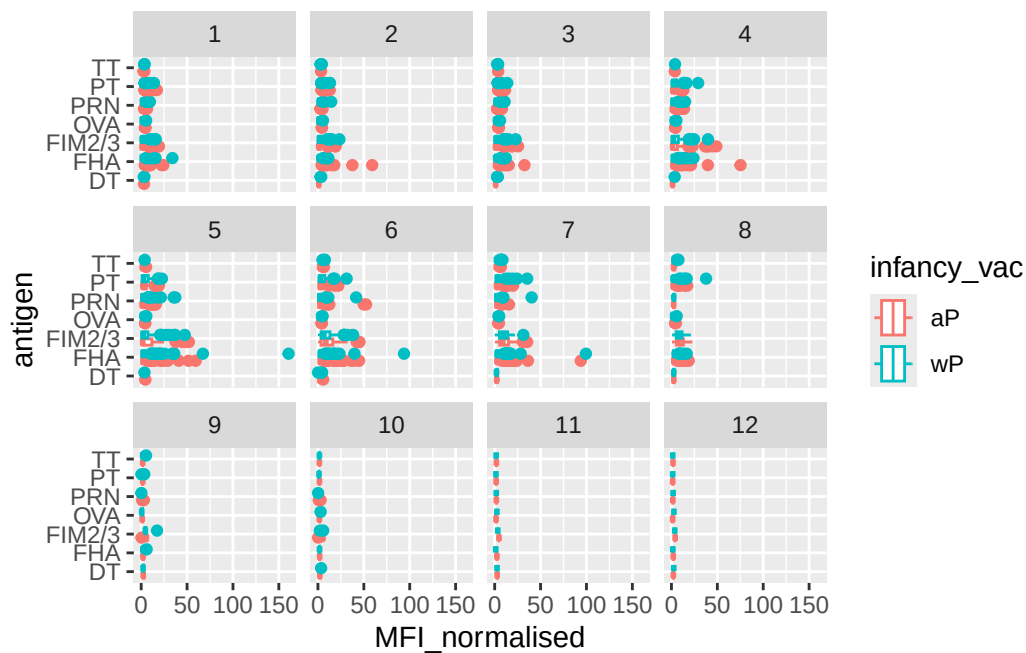


```
ggplot(igg) +
  aes(MFI_normalised, antigen,) +
  geom_boxplot() +
  facet_wrap(~infancy_vac)
```



Is there a difference with time (I.e. before booster shot vs. after booster shot)

```
ggplot(igg) +
  aes(MFI_normalised, antigen, col=infancy_vac) +
  geom_boxplot() +
  facet_wrap(~visit)
```

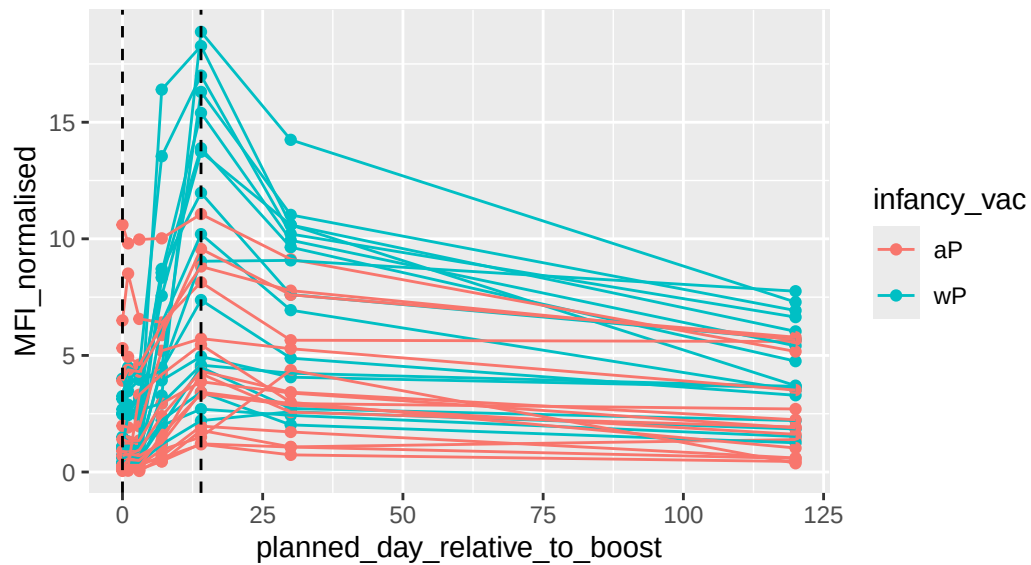


```
abdata.21 <- abdata %>% filter(dataset == "2021_dataset")

abdata.21 %>%
  filter(isotype == "IgG", antigen == "PT") %>%
  ggplot() +
    aes(x=planned_day_relative_to_boost,
        y=MFI_normalised,
        col=infancy_vac,
        group=subject_id) +
    geom_point() +
    geom_line() +
    geom_vline(xintercept=0, linetype="dashed") +
    geom_vline(xintercept=14, linetype="dashed") +
    labs(title="2021 dataset IgG PT",
         subtitle = "Dashed lines indicate day 0 (pre-boost) and 14 (apparent peak levels)")
```

2021 dataset IgG PT

Dashed lines indicate day 0 (pre-boost) and 14 (apparent peak levels)



Q18. Does this trend look similar for the 2020 dataset?

Yes, the trend looks similar in the 2020 dataset. IgG antibody levels against PT increase after the booster vaccination, peak around day 14, and then gradually decline, which is consistent with a typical immune response.